

Review

Does plyometric training improve strength performance? A meta-analysis

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Received 28 May 2009; received in revised form 4 August 2009; accepted 12 August 2009

Abstract

Majority of the research suggests plyometric training (PT) improves maximal strength performance as measured by 1RM, isometric MVC or slow velocity isokinetic testing. However, the effectiveness of PT depends upon various factors. A meta-analysis of 15 studies with a total of 31 effect sizes (ES) was carried out to analyse the role of various factors on the effects of PT on strength performance. The inclusion criteria for the analysis were: (a) studies using PT programs for lower limb muscles; (b) studies employing true experimental design and valid and reliable measurements; (c) studies including sufficient data to calculate ES. When subjects can adequately follow plyometric exercises, the training gains are independent of fitness level. Subjects in either good or poor physical condition, benefit equally from plyometric work, also men obtain similar strength results to women following PT. In relation to the variables of program design, training volume of less than 10 weeks and with more than 15 sessions, as well as the implementation of high-intensity programs, with more than 40 jumps per session, were the strategies that seem to maximize the probability to obtain significantly greater improvements in performance ($p < 0.05$). In order to optimise strength enhancement, the combination of different types of plyometrics with weight-training would be recommended, rather than utilizing only one form ($p < 0.05$). The responses identified in this analysis are essential and should be considered by the strength and conditioning professional with regard to the most appropriate dose–response trends for PT to optimise strength gains.

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Keywords: Force; Effect size; Lower limb; Training volume; Intensity

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1. Introduction

Muscular strength and power are considered as critical elements for a successful athletic performance, as well as for carrying out daily activities and occupational tasks.^{1,2}

Much research has been focused on the development of maximal strength performance as this neuromuscular quality appears to underpin most other domains of human physical capacity. Although various training methods, including weight-training,^{3,4} explosive and ballistic-type resistance training methods,⁵ electrostimulation training,^{6,7} and vibration training⁸ have been effectively used for the enhancement of strength performance, there is solid research evidence that

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plyometric training (PT) is also effective for improving ballistic and maximal strength.^{9–13}

Plyometrics refers to exercises that are designed to enhance neuromuscular performance. For the lower body this involves application of jump, hopping and bounding training. Plyometric exercises constitute a natural part of most sport movements as they involve jumping, hopping and skipping (i.e., such as high jumping, throwing or kicking).^{14–16} Plyometric exercises are implemented in various forms depending on the purpose of the training program. Typical plyometric exercises include the countermovement jump (CMJ), the drop jump (DJ) and the squat jump (SJ). These exercises can either be combined within a training program or can be applied independently. Furthermore, plyometrics can be performed at various intensity levels, ranging from low-intensity double-leg hops to high-intensity unilateral drills. As far as the lower body is concerned, plyometrics includes the performance of various types of body-weight jumping exercises, such as DJ, CMJ, alternate-leg bounding, hopping, and other stretch-shortening cycle (SSC) exercises.^{17–20} These exercises are characterised by SSC actions, that is, they start with a rapid stretch of a muscle (eccentric phase) and are followed immediately by a rapid shortening of the same muscle (concentric phase).^{17,19,20–23}

Research indicates PT improves strength, power output, coordination, and athletic performance.^{24–28} Numerous studies on PT have demonstrated improvements in maximal strength,^{9–13,29} ranging from 11 kg to 60 kg (performing exercises such as DJ, CMJ, SJ, combined jumps or combined weights and plyometric training) that could be attributed to the enhanced coordination and the individual's ability to rapidly increase muscle tension resulting in greater maximal rate of force development (RFD).^{11,37} In addition, a number of authors determined^{24,30–35} significant positive effects of PT on maximal strength when compared with other training modalities (i.e., weight-training, eccentric training, isometric training). However, several authors have shown that for optimising maximal strength enhancement, the combination of training modalities (i.e., plyometrics and high-intensity resistance training) is recommended rather than using only a single modality.^{9,12,16} However, the characteristics of a training program that achieves better gains are not clear.

The effects of PT may differ depending on the various subject characteristics, such as training level,^{36–38} gender,³⁹ age,^{40–42} sport activity or familiarity with plyometric training.^{11,29} Research studies that combine these variables in different ways sometimes lead to conflicting results.^{7,12,43} Other factors that seem to determine the effectiveness of PT are program duration and training volume. Research studies have used numerous combinations of duration, intensity and volume characteristics^{4,10,12,13,44,45} therefore, the optimal combination of these factors for maximum enhancement remains unclear.

Despite the advantages of PT, the principal issue of determining the optimal elements of a plyometric program remains inconclusive. Identification of the role of the various ele-

ments of a PT program with regard to their effectiveness can be achieved with the use of meta-analysis: a method that overcomes the problems both of small sample size and low statistical power. Meta-analysis is a quantitative approach in which individual study findings addressing a common problem are statistically integrated and analysed.⁴⁶ Because meta-analysis can effectively increase the overall sample size, it can also provide a more precise estimate of the effect of PT on strength performance. In addition, meta-analysis can account for the factors partly responsible for the variability in treatment effects observed among different training studies.^{4,9,11–13} Thus, the purpose of this study was to examine the influence of various factors on the effectiveness of PT using a meta-analysis approach.

2. Methods

A search was performed using key words in the English and French languages (e.g., jump training, drop jump, depth jump, stretch-shortening cycle, plyometric, plyometrics, training of power, plyometric training, pliometrique, and entrainement pliometrique). These key words were applied in the databases ADONIS, ERIC, SPORTSDiscus, EBSCOhost, MedLine and PubMed. Moreover, manual searches of relevant journals and reference lists obtained from articles were conducted. The present meta-analysis includes studies published in journals that have presented original research data on healthy human subjects. No age, gender or language restrictions were imposed during the search stage.

Research studies implementing PT programs for lower-limbs were used. Investigations involving training of the upper-limbs as well as summaries or abstracts were rejected. A total of 25 studies were initially identified.

The next step was to select studies with respect to their internal validity. Selection was based on the recommendations by Campbell and Stanley⁴⁷ and included; (1) studies involving a control group, (2) randomised control studies, (3) studies using instruments with high reliability and validity, (4) studies with minimal experimental mortality. Fifteen studies were selected after having completed all quality conditions (Table 1).^{4,7,9,10–13,29,33,37,43–45,48,49}

Each study was read and coded independently by 2 investigators using different moderator variables. Because of the high number of variables that may affect training effectiveness, independent variables were grouped into the following areas: (1) subject characteristics: variables included age (years), body mass (kg), height (cm), previous experience, group size, level of fitness, sports level and type of sport activity; (2) program exercises: variables included combination with other types of exercise, intensity of session, type of plyometric exercises and resistance; (3) program elements: variables included frequency of weekly sessions, program duration, drop height, number of jumps per session, number of exercises per session and rest intervals between series of exercises; and (4) outcome measurements: the type of

Table 1

Summary of characteristics of all studies meeting the inclusion criteria.

Authors	Gr	Year	Treatm	n	G	Age	W	H	Exp	KPly	Fit	MFG	ES	Freq	D wk	Int	BH	NoJ	NoE	Tply	R	JL	Test
Blakey et al. ⁹	E	1987	Ply + WT	31	M	19.5	–	–	NR	Y	N	11.43	0.37	3	8	Mod	40	30	1	DJ	–	NR	1RM PRESS
Buckley et al. ¹⁰	E	2003	Ply + WT	26	M	23.5	–	–	NR	N	N	65	1.3	3	8	Low	–	–	3	C	–	N	1 RM PRESS
Clutch et al. ¹¹	E	1983	Ply + WT	12	M	20.9	77.7	179	NA	N	N	18.56	1.5	2	4	Low	–	40	1	SJ	120	N	1RM SQUAT
Clutch et al. ¹¹	E	1983	Ply + WT	12	M	20.9	77.7	179	NA	N	N	10.59	0.79	2	4	Low	30	40	1	DJ	120	N	1 RM SQUAT
Clutch et al. ¹¹	E	1983	Ply + WT	12	M	20.9	77.7	179	NA	N	N	14.58	1.22	2	4	Low	75	40	1	DJ	120	N	1RM SQUAT
Fatouros et al. ¹²	E	2000	Plyo	41	M	21.1	83.4	178	NA	N	B	16.4	1.3	3	12	H	30	150	5	C	–	N	1 RM SQUAT
Fatouros et al. ¹²	E	2000	Ply + WT	41	M	20.1	79.9	178	NA	N	B	36.1	2.05	3	12	H	30	150	5	C	–	Y	1RM SQUAT
Fatouros et al. ¹²	C	2000	–	41	M	20.5	80.8	181	NA	–	B	1.7	0.07	–	–	–	–	–	–	–	–	–	1RM SQUAT
Fowler et al. ²⁹	E	1995	Ply + WT	18	M	22.7	77.5	181.5	Reg	Y	G	17	0.97	4	3	H	–	50	2	SJ	–	Y	1 RM SQUAT
Fry et al. ⁴⁸	E	1991	Ply + WT	14	F	19.6	64.3	171.9	Nat	N	E	14.1	1.11	2	12	Mod	–	–	7	C	–	N	1RM SQUAT
Herrero et al. ⁴⁴	E	2006	Plyo	40	M	20.8	79.7	179	NA	N	N	3.99	0.13	2	4	H	–	100	–	C	180	N	ISOMETRIC
Herrero et al. ⁴⁴	E	2006	Ply + EMS	40	M	21.4	80.2	179	NA	N	N	40.1	1.69	4	4	H	–	100	–	C	180	N	ISOMETRIC
Kramer et al. ³⁷	E	1983	Ply + WT	28	F	21.3	66.5	170	Nat	Y	E	48	0.81	3	9	Mod	30	60	4	S + D	–	N	1RM SQUAT
Lyttle et al. ³³	E	1996	Plyo	39	M	23.9	79.1	182.3	Reg	N	G	14	0.57	2	8	Mod	–	40	1	SJ	–	N	1RM SQUAT
Lyttle et al. ³³	E	1996	Ply + WT	39	M	23.8	72.5	178.4	Reg	N	G	15	0.72	2	8	Mod	20	10	1	DJ	–	N	1RM SQUAT
Lyttle et al. ³³	C	1996	–	39	M	20.6	74.2	177.3	Reg	–	G	2.6	0.06	–	–	–	–	–	–	–	–	–	1RM SQUAT
Maffiuletti et al. ⁷	E	2002	Ply + EMS	20	M	21.8	80.5	190.7	Reg	Y	E	20.3	1.11	3	4	Mod	40	50	2	C + D	180	N	ISOKINETIC
Maffiuletti et al. ⁷	C	2002	–	20	M	22.3	75.2	180.6	Reg	–	E	5.6	0.17	–	–	–	–	–	–	–	–	–	ISOKINETIC
Martel et al. ⁴⁵	E	2005	Ply + Sw	19	F	15	64	167	Nat	N	G	9.5	0.41	2	6	Mod	–	–	–	C	30	N	ISOKINETIC
Martel et al. ⁴⁵	C	2005	–	19	F	14	57	164	Nat	–	G	5	0.11	–	–	–	–	–	–	–	–	–	ISOKINETIC
Polhemus et al. ⁴⁹	E	1980	Ply + WT	29	M	25	–	–	NR	Y	G	20.08	0.66	3	6	H	45	50	3	DJ	60	N	1RM SQUAT
Polhemus et al. ⁴⁹	E	1980	Ply + WT	32	M	25	–	–	NR	Y	G	29.37	1.38	3	6	H	45	50	3	DJ	60	Y	1RM SQUAT
Sáez-Sáez de Villarreal et al. ¹³	E	2008	Plyo	42	M	22.3	75.6	174.7	NA	N	N	21.5	0.67	1	7	H	60	60	1	DJ	60	N	1 RM PRESS
Sáez-Sáez de Villarreal et al. ¹³	E	2008	Plyo	42	M	23.1	80.1	176.6	NA	N	N	29.59	0.75	2	7	H	60	60	1	DJ	60	N	1RM PRESS
Sáez-Sáez de Villarreal et al. ¹³	E	2008	Plyo	42	M	21.8	72.68	175.5	NA	N	N	50.56	1.62	4	7	H	60	60	1	DJ	60	N	1RM PRESS
Sáez-Sáez de Villarreal et al. ¹³	C	2008	–	42	M	23.6	78.56	180.3	NA	–	N	16	0.28	–	–	–	–	–	–	–	–	–	1RM PRESS
Wilson et al. ⁴	E	1993	Plyo	13	M	22.1	71.6	174	NR	Y	N	1.3	0.02	2	10	Mod	20	60	2	DJ	180	N	ISOMETRIC
Wilson et al. ⁴	C	1993	–	13	M	24.1	76.1	173	NR	–	N	–6.22	–0.03	–	–	–	–	–	–	–	–	–	ISOMETRIC
Witzke et al. ⁴³	E	2000	Plyo	25	F	14.6	61.2	164.3	Reg	N	N	12	0.55	3	36	Mod	–	120	6	C	–	N	ISOKINETIC
Witzke et al. ⁴³	C	2000	–	28	F	14.5	61	165.1	Reg	–	N	5.1	0.15	–	–	–	–	–	–	–	–	–	ISOKINETIC

Treatm (treatment), Plyo (plyometric), Ply + WT (weight-training), Ply + EMS (electrostimulation), Ply + Sw (in water)/JL (jump loaded): Y (yes), N (no), NR (not reported). Gr (group): E (experimental)/G (gender): M (male), F (female)/W (weight (kg)/H (height (cm)/Kply (knows plyometric) N (no), Y (yes)/test: Performance test. Exp (experience): Nat (national), Reg (regional), NA (no athlete) NR (not reported)/Fit (fitness): E (elite), G (good), N (normal), B (bad). Tply (type of plyometric): C (combined), S + D (SJ + DJ), C + D (CMJ + DJ), DJ, SJ/R (rest) (sec)/MFG (kg) (maximal force gains). Freq (frequency Days/wk)/D wk (duration weeks)/Int (intensity): H (high), Mod (moderate) Low/BH (box height (cm))/NoJ (number of jumps)/NoE (number of exercises).

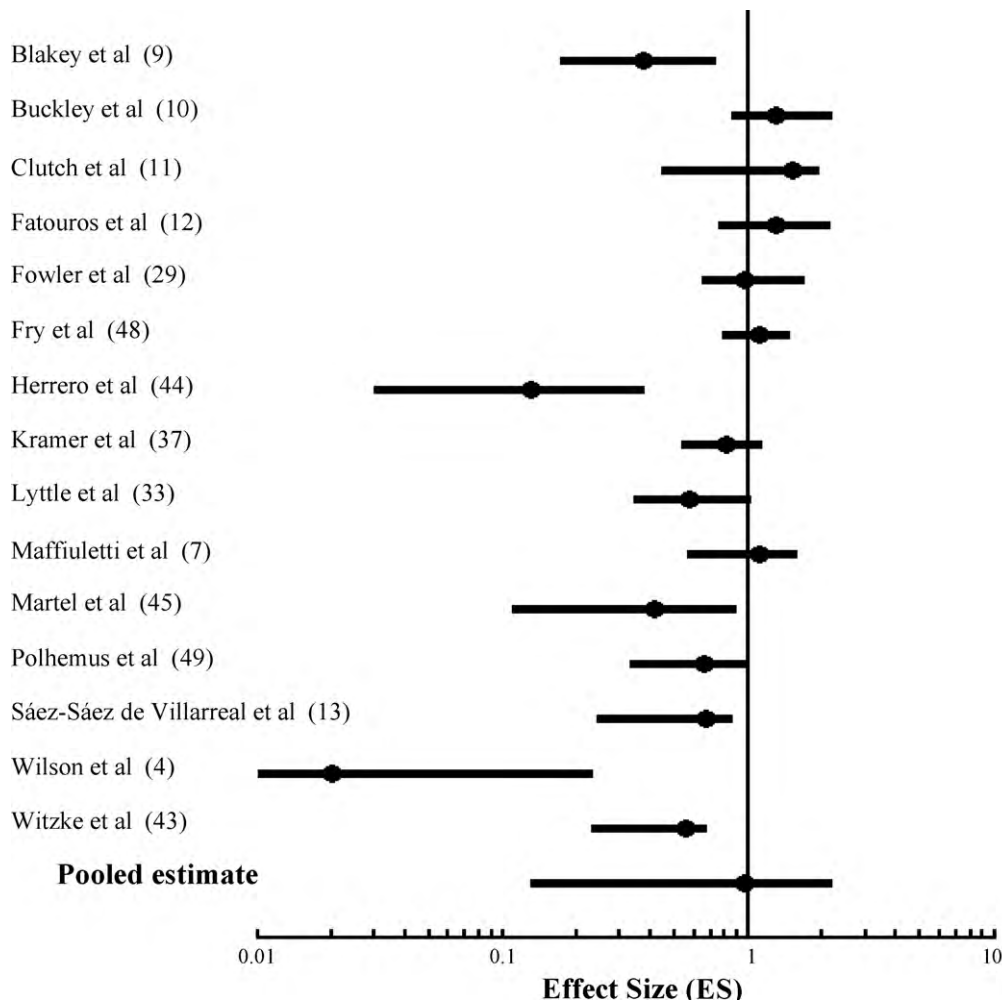


Fig. 1. Effect size (ES) of all studies meeting the inclusion criteria. Horizontal bars represent 95% confidence intervals.

strength test used to identify gains (1RM squat, isokinetic, isometric and 1RM leg press). The coding agreement between investigators was determined by dividing the variables coded the same by the total number of variables. A mean agreement of 0.90 is accepted as an appropriate level of reliability in the coding procedure.⁵⁰ Mean agreement was 0.94 in our study. Each coding difference was scrutinised by both investigators and was resolved before the analysis.

The ES is a standardised value that permits the determination of the magnitude of the differences between the groups or experimental conditions.⁵¹ Gain ESs were calculated using Hedges and Olkin's g ,⁴⁶ using formula (1):

$$g = \frac{M_{\text{post}} - M_{\text{pre}}}{SD_{\text{pooled}}} \quad (1)$$

where M_{post} is the mean for the posttest and M_{pre} is the mean for the pretest, and SD_{pooled} is the pooled SD of the measurements (2):

$$SD_{\text{pooled}} = \frac{(M_{\text{post}} - M_{\text{pre}})}{\sqrt{((n_1 - 1) \cdot SD_1^2 + (n_2 - 1) \cdot SD_2^2) / (n_1 + n_2 - 2)}} \quad (2)$$

It has been suggested,^{51–53} that ES should be corrected for the magnitude of sample size of each study. Therefore, correction was performed using formula (3):

$$\frac{1 - 3}{4m - 9} \quad (3)$$

where $m = n - 1$, as proposed by Hedges and Olkin.⁴⁶

3. Statistical analysis

To examine the effect of the categorical independent variables on the ES, an analysis of variance (ANOVA) was used.^{52,54,55} In the case of quantitative independent variables (e.g. age, height, duration of the treatment in weeks, number of repetitions per session) a Pearson's (r) correlation test was used to examine the relationships between ESs and

Table 2

Analysis for independent variables of subject characteristics.

Independent variables	Average (kg) $\pm$ SD	<i>F</i>	Level	ES	SD	<i>n</i>	<i>r</i>
Subject characteristics							
Age (y)						24	0.242
Body mass (kg)						19	0.451*
Height (cm)						19	0.396
Group size						24	0.05
Previous experience		<i>F</i> (1,24)=0.05	<i>p</i> =0.944				
Familiarized	25.27 $\pm$ 18.91			0.90	0.58	8	
Not familiarized	23.72 $\pm$ 16.34			1.39	0.95	16	
Fitness		<i>F</i> (3,24)=1.27	<i>p</i> =0.304				
Bad	26.25 $\pm$ 13.93			1.67	0.53	2	
Normal	24.24 $\pm$ 18.59			0.89	0.55	12	
Good	21.65 $\pm$ 18.34			0.98	0.41	7	
Elite	28.52 $\pm$ 14.88			0.81	0.35	3	
Gender		<i>F</i> (1,24)=1.18	<i>p</i> =0.333				
Male	24.87 $\pm$ 17.05			1.01	0.62	20	
Female	20.91 $\pm$ 18.16			0.72	0.30	4	
Sport level		<i>F</i> (2,19)=1.19	<i>p</i> =0.329				
National	25.82 $\pm$ 17.60			0.63	0.58	3	
Regional	14.07 $\pm$ 4.10			0.64	0.31	6	
No athletes	24.19 $\pm$ 14.54			1.17	0.58	10	
Sport activity		<i>F</i> (6,19)=0.655	<i>p</i> =0.687				
Volleyball	14.63 $\pm$ 5.41			0.87	0.40	2	
Basket	7.19 $\pm$ 2.56			0.41	–	1	
Body building	13.21 $\pm$ 2.99			0.80	0.49	6	
Rowing	48.23 $\pm$ 7.32			0.80	–	1	
Swimming	20.23 $\pm$ 9.93			0.50	–	1	
Phys. Ed. Stud.	27.12 $\pm$ 16.69			0.97	0.59	6	
American football	49.72 $\pm$ 13.56			0.80	0.51	2	

* $p < 0.05$; ES: effect size; SD: standard deviation; *n*: sample; level: alpha level; *r*: Pearson's correlation coefficient; *p*: alpha level.

variable values.⁵² Statistical significance was set at $p \leq 0.05$ for all analyses. The scale used for interpretation was the one proposed by Rhea,⁵⁶ which is specific to strength training research and the training status of the subjects to evaluate the relative magnitude of an ES. The magnitudes of the ESs were considered either trivial (<0.35), small (0.35–0.80), moderate (0.80–1.50) or large (>1.5) (Fig. 1).

4. Results

The analysis revealed that the average ES of the PT group (0.97; $n = 24$; 24.25 kg) was significantly higher ($p < 0.05$) compared to the ES of controls (0.11; $n = 7$; 4.25 kg).

With regards to the subject characteristics, the results indicate a significant correlation coefficient for body mass ($r = 0.451$) with the magnitude of the ES. However, there was no significant correlation coefficient for age ($r = 0.242$), height ($r = 0.396$) or group size ($r = 0.05$), with the magnitude of the ES (Table 2). Results of the ANOVA comparisons were no significant effects ($p > 0.05$) in any of the variables measured (i.e., previous experience, fitness level, gender, sport level and sport activity).

There was a significant effect regarding the intensity of session and different combinations of PT ($p < 0.05$). No dif-

ferences in ESs were found among the type of plyometric exercises or among programs with or without added resistance (Table 3).

There was a positive relationship ($p < 0.05$) between the frequency of sessions per week ($r = 0.439$) with PT ES, but no significant effects were found between program duration (wk) ($r = -0.218$), drop height (cm) ($r = 0.031$), number of repetitions per session ($r = -0.223$) and number of exercises per session ($r = -0.152$) with the PT ES (Table 4). No differences in ES ($p > 0.05$) were found among the different strength tests (Table 5).

5. Discussion

The results of this investigation support numerous previous studies^{4,9,11,13,33,43} that have concluded that PT is an effective training method for the improvement of strength performance (ES=0.97; i.e., plyometric group). Thus, the reported strength gains of >20 kg resulting from PT could be of practical relevance for trained athletes in sports aiming at achieving optimum strength performance. The present meta-analysis offers robust quantitative evidence for this conclusion and provides some valuable information concerning the importance of controlling

Table 3

Analysis of variance results on the differences of ES between various elements of plyometric training independent variables of program elements.

Independent variables	Average (kg) $\pm$ SD	<i>F</i>	Level	ES	SD	<i>n</i>
Program exercises						
Combination with other types of exercise		$F(3,24) = 3.383$	$p = 0.035^*$			
Plyometric	19.99 $\pm$ 13.93			0.64	0.48	8
Ply + weight-training	27.76 $\pm$ 19.53			1.21	0.57	13
Ply + electrostimulation	30.27 $\pm$ 14.07			1.42	0.41	2
Ply in water	9.56 $\pm$ 0.41			0.41	–	1
Intensity of session		$F(2,24) = 8.98$	$p = 0.006^*$			
High	31.46 $\pm$ 17.02			1.32	0.69	10
Moderate	17.27 $\pm$ 11.54			0.61	0.30	10
Low	27.18 $\pm$ 25.42			1.20	0.29	4
Types of plyometric exercises		$F(4,24) = 1.03$	$p = 0.425$			
Combined	25.43 $\pm$ 19.48			0.97	0.67	8
Squat jump	16.52 $\pm$ 2.31			1.01	0.46	3
Drop jump	25.16 $\pm$ 17.41			1.09	0.64	11
SJ + DJ	27.59 $\pm$ 18.85			0.61	–	1
CMJ + DJ	20.31 $\pm$ 12.32			1.11	–	1
Resistance		$F(1,23) = 1.32$	$p = 0.263$			
Added weight	37.49 $\pm$ 21.21			1.29	0.65	3
Weightless	23.39 $\pm$ 16.54			0.94	0.47	20

* $p < 0.05$; SJ: squat jump; CMJ: countermovement jump; DJ: drop jump; Ply: plyometric; ES: effect size; SD: standard deviation; *n*: sample; level: alpha level.

Table 4

Pearson's correlation coefficients (*r*) between various program elements and training gains.

Training program variables	<i>n</i>	<i>r</i>	<i>p</i>
Frequency session/week	26	0.439	0.05*
Program duration (wk)	26	–0.218	
Drop height (cm)	16	0.031	
Number of jumps per session	23	–0.223	
Number of exercise/session	23	–0.152	
Rest between sets (s)	13	–0.243	

* $p < 0.05$; *n*: sample, *r*: Pearson's correlation coefficient; *p*: alpha level.

some determinant variables for the improvement of the performance.

Some authors suggest that PT requires appropriate technical ability as well as sufficient levels of muscle strength and joint coordination.^{57,58} However, Wilson et al.,⁵⁹ report that improvement in performance from PT is not determined by initial strength level. Similarly, the results of the present meta-analysis indicate lower but not significantly different ESs for more experienced subjects and with good or excellent fitness levels in comparison with less experienced subjects and with poor fitness (Table 2). These results might indicate that

when subjects can perform plyometric exercises with adequate technique, the training gains are independent of fitness level. However, it is known that when less fit people start exercising regularly, they could achieve higher gains during the first weeks of training in comparison with well-trained individuals measured by most of the indices of physical fitness.⁶⁰ Furthermore, a major part of the improvements in untrained subjects during the initial weeks in ballistic-type strength training is probably due to adaptations of the neural system, such as increased motor unit firing frequency, improved motor unit synchronisation, increased motor unit excitability, and increase in efferent motor drive. Also, a reduction of the antagonist and an improved co-activation of the synergist muscles may explain part of the changes.⁶¹ In a study of Aagaard et al.,⁶² the major component of the training induced improvements after 14 weeks of resistance training, were explained by increases in efferent neural drive. This may be one explanation for the higher changes in less experienced individuals.

An interesting finding of this study was that men demonstrated similar gains compared with women (Table 2). However, the large difference in sample size between men

Table 5

Analysis for independent variables of outcome measurement.

Independent variables	Average (kg) $\pm$ SD	<i>F</i>	Level	ES	SD	<i>n</i>
Outcome measurement						
Strength test		$F(3,24) = 1.22$	$p = 0.325^*$			
1RM Squat	24.90 $\pm$ 18.6			1.11	0.57	15
Isokinetic	18.37 $\pm$ 10.06			0.57	0.38	3
Isometric	19.03 $\pm$ 18.79			0.67	0.58	3
1RM leg press	33.88 $\pm$ 14.99			1.01	0.52	3

* $p < 0.05$; ES: effect size; SD: Standard deviation; *n*: sample; level: alpha level.

and women and the small number of ESs available may account for this observation. The reasons for this similarity are not clear. Muscle strength (an absolute value) of women is equivalent to 50–60% of men when we compare isometric muscle strength among men and women.^{63,64} Furthermore, gender difference was found in SSC ability, and the ability was superior in men as compared with women. The ability to use SSC in women was 64.1% that of men. It was reported that the dominance in women of type I muscle fibers⁶⁵ and a difference in the degree of inhibition in the nervous system⁶⁶ may be related to the gender difference in muscle strength. In addition, muscular morphologic characteristics (muscle fascicle length and pennation angles) may be influential. However, when strength is expressed relative to muscle cross-sectional area, no significant difference exists between sexes, which indicate that muscle quality (peak force per cross-sectional area) is not sex specific.^{64,67} According to previous studies,^{68,69} it was reported that the ability to use elastic energy following eccentric muscle action is superior in women vs. men. Komi and Bosco²³ have pointed out that the ability to endure extension load is superior in men compared with women, but the ability to use elastic energy is inferior in men. Furthermore, Aura and Komi⁶⁸ report that women are superior in storage and recycling ability of elasticity energy compared with men when the extension load is small, but inferior when the load is large. This may depend on a gender difference of muscle stiffness and inhibition of the central nervous system.⁶⁹

In the present study, strength improvements are significantly higher when plyometrics are combined with other types of exercises (i.e., plyometric + weight-training and plyometric + electrostimulation) (Table 3). The differences might be attributed to several reasons. First, the nature of the training protocol, the type of plyometric and weight-training exercises (i.e., full-squats, parallel squats, Olympic exercises, etc) used and second, the training stimulus. There is a possibility that the subjects in the combination training group were exposed to a higher training stimulus than those in the other groups, that is, the total workload was not equated between groups. It would be very interesting if future studies made an attempt to equate workloads between groups when comparing different training methods. Another difference is the model used to provide the training stimulus to subjects. Training intensity, volume, and exercise selection followed the principle of progressive overload, starting with lower intensities, single-joint exercises, and less complex exercise techniques, and progressing to higher intensities, multi-joint exercises, and more complex techniques. In any case, the optimal training strategy to enhance dynamic athletic performance appears to be a hybrid between traditional weight-training and PT. That is, strength performance gains will be optimised by the use of plyometric + weight-training at a training load that maximises the mechanical force output of the exercise. Hence, the combined group tended to perform better in activities of maximal force. This may be due to the fact that this combination of exercises may better

facilitate the neural and mechanical mechanisms that enhance performance in activities of maximal force.

The results of this investigation suggest that when the intensity is high during the session, there is a greater improvement in strength performance (Table 3). Some authors^{40,70–72} determined that performance is higher during DJs, followed by CMJs and then SJs. This is mainly attributed to the different characteristics of movement and, thus, to the different utilisation of SSC characteristics. For these reasons, the combination of various exercises may result in higher gains compared with the performance of each exercise alone. However, the present results show that a combination of SJs, CMJs, and DJs demonstrates similar ES compared with the use of a single type of exercise (Table 3). The specific effects of PT on strength performance in the different types of vertical jumps could be of particular importance. It has been suggested that PT is more effective in improving performance because it enhances the ability of subjects to use the elastic and neural benefits of the SSC.³⁹ This could also be attributed to differences in the use of SSC characteristics.^{23,40} A SJ mainly consists of a concentric (push-off) phase, whereas a CMJ involves an eccentric and concentric phase.¹⁸ The results of our study do not support these suggestions. Specifically, our data indicate that PT produces similar positive effects whether fast SSC jumps (i.e., DJ) or concentric-only jumps (i.e., SJ), or even slow SSC jumps (i.e., CMJ), are used. That is, all the treatments increased strength performance (i.e., 1RM squat, isometric, isokinetic or 1RM leg press). This agrees with previous results^{11,37} that have shown that PT enhances an individual's ability to rapidly develop force. Performing PT involves the rapid development of maximal force during the eccentric phase of motion. It has been previously reported that the body experiences tremendous impact forces during foot contact with the ground in vigorous locomotion,^{73–77} thus, one may speculate that muscle force stimulus during any PT (i.e., DJs, CMJs, and combined) can be effective for strength development.

Some research studies have shown that PT with additional weights (vests, barbell on the back, etc.) demonstrated higher gains.^{12,29,49} In addition, Wilson et al.,⁴ clearly showed that jumping with a barbell and traditional resistance training were far superior for increasing maximal strength compared to plyometrics. However, the results of the meta-analysis indicated no significant differences among the training conditions (Table 3). This suggests in some cases, that using additional weights in training could not cause significant gains in performance. It could be suggested, then, that training with additional loads might increase not only resistance, but also contact time. However, the longer the contact time, the less effective the SSC.²⁷ Therefore, superior training effects using additional weights can not be guaranteed.

Volume and frequency are very important parameters to be taken into account for an optimum PT program design. Our analysis suggests that training for less than 10 weeks (i.e., between 6 and 10) with 3 sessions per week is more beneficial than similar programs of longer duration. Simi-

larly, treatment with more than 15 sessions increases strength performance, whereas performance of more than 40 repetitions per session seemed to be the most beneficial volume (Table 4). However, in agreement with previous studies¹³ a short-term PT program with a moderate training frequency and volume of jumps (2 d wk⁻¹, 840 jumps per week), produced similar enhancements in strength performance but greater training efficiency (number of jumps/% of the improvement) compared with high training frequency (4 d wk⁻¹, 1680 jumps per week). Conceptually taken on the whole, the present data would indicate that increasing the number of jumps in previously moderately trained men does not seem to be the best stimulus for improving strength performance during short-term training periods compared with high jump-training volumes. These results also suggest that there is a maximum training volume threshold over which further increases in volume are no longer advantageous.

6. Conclusion

In conclusion, the present study demonstrates that PT significantly improves strength performance. The estimated improvements in strength as a result of PT could be considered as practically relevant—for example, an improvement in strength of >20 kg (i.e., ES=0.97) could be of high importance for trained athletes in sports relying on strength performance. According to our results, when subjects can perform plyometric exercises with adequate technique, the training gains are independent of fitness level. On the other hand, subjects of both high and lower physical condition benefit equally from PT, although men obtain similar strength results than women after PT. A training volume of less than 10 weeks (with more than 15 sessions) using high intensities (with more than 40 jumps per session) is the strategy that will maximise the probability of obtaining significant improvements in performance. It is also probable that there is a threshold training volume threshold over which further increases in volume may no longer be advantageous. Another important conclusion is that it is more beneficial to combine plyometrics with weight-training than to utilise only the single modality.

Practical implications

- The effects of PT may vary because of a large number of variables, such as training programme design, subject characteristics (gender, age), training level, the specific sport activity, familiarity with PT, program duration, and training volume or intensity. These variables should be taken into account by strength and conditioning professionals, who must consider the most appropriate PT approach based on the fundamental movement patterns, technique, volume, frequency, intensity, energy system

requirements, and potential injury analysis for a given sport.

- For an individual athlete, initial training status and training experience must be considered, and specific fitness limitations should be stressed. The strength and conditioning coach may consider taking into account the dose–response trends identified in this analysis to prescribe the appropriate level of training.

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